

Jerry (Yuze) Li

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EDUCATION

University of Waterloo

Bachelor of Computer Science | Schulich Leader Scholarship (\$100,000)

Expected 2029

Waterloo, Canada

EXPERIENCE

Mathos AI (YC W24) *Software Engineer Intern*

June 2026 – Present

- Shipped MP4 export for AI math videos, rendering Remotion jobs in isolated AWS Fargate sandboxes via Celery.
- Built an OAuth 2.0 and MCP server so ChatGPT and Claude users can generate and download Mathos videos in chat.
- Shipped lifecycle emails on AWS SES and EventBridge with engagement tracking and Gmail reply threading.
- Rebuilt the AI tutor's step grading on a fast judge model plus SymPy checks, sharply improving accuracy and speed.

TopsOJ *Founder & Lead Developer*

July 2023 – Present

- Built a competitive-math judge with 100,000+ users in 180+ countries (Google Analytics) and 128,000+ submissions.
- Designed EloMMR ratings; led a 30-person team across engineering, contest ops, and sponsors like 3Blue1Brown.

Quoted Tech Computers *Software Engineer Intern*

July 2025 – Aug. 2025

- Automated barcode-label generation and CUPS printing with Node.js/Express, cutting manual processing time by 70%.
- Built Playwright/Selenium scrapers that structure competitor catalogs for pricing and inventory analysis.

RESEARCH

Computational Astrophysics

Nov. 2024 – Mar. 2026

Sole author of three papers, supervised by Dr. Alexander Mushtukov (University of Oxford)

Remote

- “Ballistic Mass Transfer Necessarily Contracts Orbits in Binary Star Systems,” *Research Notes of the AAS* (2026).
Showed the orbit loses angular momentum as $1 - f$ and shrinks at every mass ratio.
- “On the Conditions Under Which Instantaneous Mass-Transfer Mappings Reproduce Conservative Limits,” *Journal of High School Science* (2026), **Best Manuscript**. A velocity-rescaling closure cut orbit error from 22% to under $10^{-4}\%$.
- “The Effective Velocity of Transferred Mass,” arXiv:2608.19107, under review at *Astronomische Nachrichten*. Unified transfer models under one parameter η that decides whether orbits widen or shrink.

PROJECTS

Cardboard Cockpit | *Godot, Python/OpenCV, ArUco, WebSockets, BLE*

Hack the North 2026

- Built most of the Godot flight sim (21 of 22 PRs): map, weapons, HUD, BLE badge cockpit, landing, and tutorial.

Daedalus | *TypeScript, Three.js/WebGL, MediaPipe*

JAMHacks 10

- Co-built a webcam-gesture 3D modeling app with MediaPipe hand tracking and Three.js; won Best Non-GenAI Hack.

Drift | *JavaScript, Chrome Extensions API*

2026

- Built a Chrome extension that opens a focused feed while ChatGPT, Claude, or Gemini think; Featured on the Web Store.

Worldline Engine | *C++, raylib, CMake*

2026

- Wrote a deterministic C++ universe simulator with seeded generation, live physics, save/replay, and automated tests.

AWARDS

USACO Platinum (1,000/1,000 on Gold) | **AIME** qualifier (3×)

AAPT PhysicsBowl Div. II: 1st in Canada, 7th globally | **Sir Isaac Newton Exam**: 4th in Canada, 9th globally

TECHNICAL SKILLS

Languages: C/C++, Python, TypeScript, JavaScript, SQL

Frameworks: Node.js, Express, React, Next.js, Celery, NumPy, SciPy, Godot, raylib, Three.js

Tools: AWS (Fargate, SES, S3), Git, Linux, SQLite, CMake, OpenCV, MediaPipe, Playwright, WebSockets